



/Users/halpern/teaching/352/labs/manual/master/rev_2022/microwave_prelab.tex

Microwaves

What is the ratio

$$\eta = \frac{E}{H} \quad (1)$$

for a plane electromagnetic wave in vacuum. Express your answer in Ω . (How?)

What is the E-field reflection coefficient for a plane wave in a material where $\eta_1 = E_1/H_1$ incident on another material where $\eta_2 \neq \eta_1$?

Just in case you want more explicit advice on what we are asking for, imagine a plane wave in region 1, let's call the E-field E_o . When it strikes the interface to region 2 there will be a reflected field in region 1, called $E_1 = \rho E_o$ and a transmitted field in region 2 with E-field $E_2 = \tau E_o$. Find how E and H match at the interface, and from that find ρ and τ .

Please make your own Smith Chart: You will plot complex reflection coefficients for various loads on a waveguide,

$$\Gamma = \frac{z - z_o}{z + z_o}$$

where z_o is the characteristic waveguide impedance and z is the complex load impedance.

Take $z_o = 50 \Omega$. Model the load as a general complex impedance, $z = r + ix$.

1. Take $r = 50 \Omega$ and $-1000 < x < +1000 \Omega$. Plot Γ with the same scaling for real (horizontal) and imaginary (vertical) components. Comment.
2. Repeat this for r equal to each of (0,100, 300) Ω , all on the same plot. Comment on the physical interpretation of the $r = 0$ curve.
3. Take $x = 50 \Omega$ and $0 < r < 1000 \Omega$. Plot Γ (on the same plot).
4. Repeat this for x in (-200,-30,0,20,100,300). Notice these are only part-circles. Why?
5. Please notice that any point inside the circle $|\Gamma| = 1$ corresponds to a unique pair of values (r, x) .